# 62670 - Exam solutions - Spring 2025

22-05-2025

## Question 1

Which of the following represents the sample space for the random experiment of measuring the waiting time (in minutes) for the arrival of a train at a station?

A. {0, … 1, … 2, … 3, …, ∞}

B. {0, 5, 10, 15, …, 60}

C. {0, 1, 2, 3, …, 59}

D. None of the above

Answer: A. The train can arrive in 0 mins, 0.5 mins, … or never (∞)

## Question 2

Consider the random experiment of throwing a dice once. Let X be the event that an odd number turns up, and Y be the event that an even number turns up. What is X U Y and X∩Y?

*Here: U represents the union and ∩ represents the intersection.*

A. X U Y = {}, X ∩ Y = {1, 2, 3, 4, 5, 6}

B. X U Y = {1, 2, 3, 4, 5, 6}, X ∩ Y = {}

C. X U Y = {}, X ∩ Y = {}

D. X U Y = {1, 3, 5}, X ∩ Y = {2}

Answer: B.

## Question 3

If $P(X|Y) = 0.97$, $P(X|Y^c) = 0.01$, $P(Y) = 0.001$, What is $P(Y|X)$?

Here $Y^c$ is the complement of $Y$.

A. 0.97

B. 0.01

C. 0.08

D. 1

Answer: C. Apply Bayes' theorem $P(Y|X) = \frac{P(X|Y)P(Y)}{P(X|Y)P(Y)+P(X|Y^c)P(Y^c)}$.

## Question 4

X is defined as the random variable that counts the number of passengers in a bus during a day, and Y is defined as the random variable that measures the waiting time for a bus at a bus stop during a day. Which of the following statements are true?

A. X and Y are discrete random variables.

B. X and Y are continuous random variables.

C. X is a continuous random variable and Y is a discrete random variable.

D. X is a discrete random variable and Y is a continuous random variable.

Answer: D.

## Question 5

For a discrete random variable, what is its probability density function?

A. It defines the probability of all possible outcomes (all values of the random variable).

B. It defines the probability of the random variable assuming the value of 1 (success).

C. It can take any real value.

D. The PDF of a discrete distribution is always a continuous function.

Answer: A.

## Question 6

Consider the following probability density function (PDF) for a random variable X:

x | 1 | 2 | 3 | 4 | 5 |

P(X = x) | 1/5 | 1/5 | 1/5 | 1/5 | 1/5 |

The mean (μ) and variance (σ2) of X is:

A. $\mu = 1, \sigma^2 = 0$

B. $\mu = 0, \sigma^2 = 1$

C. $\mu = 3, \sigma^2 = 2$

D. $\mu = 3.5, \sigma^2 = 2.92$

Answer: C. $X \sim Unif(1, 5)$ and the mean = $\mu = \frac{1+5}{2} = 3$, $\sigma^2 = \frac{n^2-1}{12} = 2$.

## Question 7

A factory produces light bulbs, and the probability that a light bulb is defective is 0.06. If a quality control inspector randomly selects 10 light bulbs, what is the probability that exactly 3 of them are defective?

Hint: Here you can consider X = "count of defective bulbs" as your random variable and proceed to identify the distribution of X.

A. 1

B. 0.997

C. 0.016

D. None of the above

Answer: C. 0.016. $X \sim Binom(n, p)$, $n = 10, p = 0.06$. Thus \$P(X = 3) = \$:

```r
dbinom(3, 10, 0.06)
```

```
## [1] 0.01680854
```

## Question 8

For a random variable X that follows a normal distribution with mean 0.5 and standard deviation 0.5, what is the value of F(1)?

Here F denotes the Cumulative Distribution Function (CDF).

A. 0.84

B. 0.48

C. 0.24

D. 0.5

Answer: A. $F(1)$ can be found by:

```r
pnorm(1, 0.5, 0.5)
```

```
## [1] 0.8413447
```

## Question 9

Given:

- X follows a normal distribution with mean 0.1 and variance 1
- Y follows a normal distribution with mean 0.1 and variance 2
- Z = 10 * X + 1 * Y

What distribution does Z follow?

A. Z follows a normal distribution with mean 0 and variance 1

B. Z follows a normal distribution with mean 102 and variance 1.1

C. Z follows a normal distribution with mean 1.1 and variance 102

D. The distribution of Z cannot be determined.

Answer: C. $E[Z] = 10 * E[X] + 1 * E[Y] = 10 * 0.1 + 1 * 0.1 = 1.1$, $Var[Z] = 10^2 * Var[X] + 1^2 * Var[Y] = 100 * 1 + 1 * 2 = 102$.

## Question 10

Given F(x) = 0.65 for a standard normal distribution. What is the value of x?

Here F denotes the Cumulative Distribution Function (CDF).

A. 0

B. 0.67

C. 0.74

D. 0.38

Answer: D. $x$ is the 0.65th quantile. It can be obtained as follows.

```r
qnorm(0.65)
```

```
## [1] 0.3853205
```

## Question 11

Given $X$ follows a Student's t distribution with degrees of freedom = 10, what is the value of x for which F(x) = 0.85?

Here F denotes the Cumulative Distribution Function (CDF).

A. 1.09

B. 0.20

C. 0.79

D. 1.6

Answer: A. $x$ is the 0.85th quantile of the Student's t distribution. It can be obtained as follows.

```r
qt(0.85, 10)
```

```
## [1] 1.093058
```

## Question 12

Which of the following about trend and seasonality of a time series is correct?

A. Trend is the range of values of the dependent variable. Seasonality is the range values of the independent variable.

B. Trend refers to the repeating patterns in the data. Seasonality refers to the systematic change in the data that is not periodic.

C. Trend refers to the systematic change in the data that is not periodic. Seasonality refers to the repeating patterns in the data.

D. None of the above

Answer: C.

## Question 13

Consider the daily stock prices of DAX, obtained by the following R commands:

```r
data(EuStockMarkets)
DAX <- EuStockMarkets[, 1]
```

Fit a Holt Winter's exponential model to DAX and predict 1 day ahead. The predicted value is:

A. 5446.39

B. 5551.73

C. 5487.64

D. None of the above

Answer: A.

```r
m <- HoltWinters(DAX)
predict(m, n.ahead = 1)
```

```
## Time Series:
## Start = c(1998, 170)
## End = c(1998, 170)
## Frequency = 260
##           fit
## [1,] 5446.39
```

## Question 14

Which of the following regarding p value and hypothesis testing is correct?

*Here H0 is the null hypothesis.*

A. The smaller the p value, the stronger the evidence against H0

B. The larger the p value, the stronger the evidence against H0

C. If p value is smaller than $\alpha$ (significance level) accept H0

D. If p value is smaller than $\alpha$ (significance level), the test is inconclusive

Answer: A.

## Question 15

Given a linear model $(f) : y = \beta_0 + \beta_1 x$

Which of the following is correct?

A. $\beta_0$ is the value of x for which $y = 0$, $\beta_1$ is the value of $\Delta x/\Delta y$, $x$ is the output variable, $y$ is the input variable

B. $\beta_0$ is the value of y for which x = 0, $\beta_1$ is the value of $\Delta y/\Delta x$, x is the input variable, y is the output variable

C. $\beta_0$ is the value of $\Delta y/\Delta x$, $\beta_1$ is the value of x for which y = 0, x is the input variable, y is the output variable

D. None of the above

Answer: B.

## Question 16

What is $(y_i - (\beta_0 + \beta_1 x_i))^2$

Here $y_i$ is the output variable's value for row (or example) i and $x_i$ is input variable's value for row (example) i, $\beta_0$ is the intercept and $\beta_1$ is the slope of a linear model.

A. The distance between a data example at row i in the data matrix and the regression line.

B. The residual

C. The error incurred by a linear model on the data for a data example at row i

D. All of the above

Answer: D.

## Question 17

Consider the following data of energy intake:

```r
energy <- c(5260, 5470, 5640, 6180, 6390, 6515, 6805, 7515, 7515, 8230, 8770)
```

Assume that the data is normally distributed. Then, the 99% confidence interval of the mean energy intake is:

A. [5986.34, 7520.92]

B. [5260, 8770]

C. [5259, 8771]

D. [5662.25, 7845.01]

Answer: D.

```r
t.test(energy, conf.level = 0.99)
```

```
## 
##  One Sample t-test
## 
## data:  energy
## t = 19.612, df = 10, p-value = 2.599e-09
## alternative hypothesis: true mean is not equal to 0
## 99 percent confidence interval:
##  5662.256 7845.017
## sample estimates:
## mean of x 
##  6753.636
```

## Question 18

For a small data sample that is normally distributed, which of the following is true?

A. The confidence interval for the mean utilizes a normal distribution, and the confidence interval for the variance utilizes a standard normal distribution

B. The confidence interval for the mean utilizes a normal distribution, and the confidence interval for the variance utilizes a Student's t distribution

C. The confidence interval for the mean utilizes a $\chi^2$ distribution, and the confidence interval for the variance utilizes a Student's t distribution

D. The confidence interval for the mean utilizes a Student's t distribution, and the confidence interval for the variance utilizes a $\chi^2$ distribution

Answer: D.

## Question 19

For the given data,

```r
energy <- c(5260, 5470, 5640, 6180, 6390, 6515, 6805, 7515, 7515, 8230, 8770)
```

which is assumed to be normally distributed, the 95% confidence interval for the standard deviation of the population is:

A. [557.5908, 3517.502]

B. [636837.5, 4017420]

C. [798.021, 2004.35]

D. None of the above

Answer: C.

```r
n <- length(energy)
sqrt(((n - 1)* var(energy)) / qchisq(1 - 0.05/2, n - 1))
```

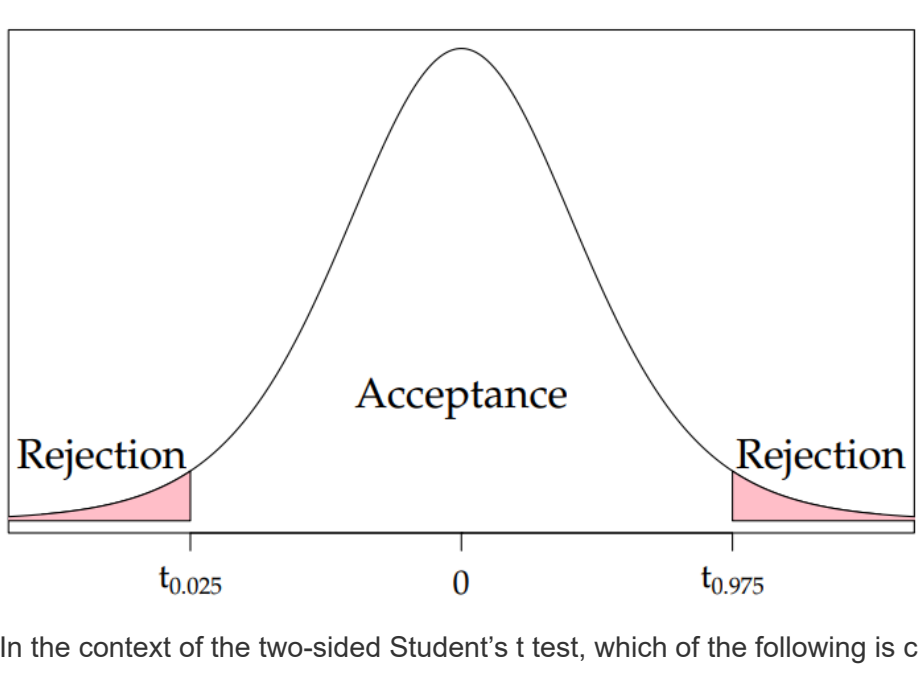
```
## [1] 798.021
```

```r
sqrt(((n - 1)* var(energy)) / qchisq(0.05/2, n - 1))
```

```
## [1] 2004.35
```

## Question 20

Consider the following plot of the Student's t distribution.



In the context of the two-sided Student's t test, which of the following is correct?

A. $t_{0.975}$ is the critical value for significance level $\alpha = 0.05$ and the 95% confidence interval is the acceptance region for the null hypothesis.

B. $t_{0.975}$ is the critical value for significance level $\alpha = 0.05$ and the 95% confidence interval is the rejection region (pink shaded region) for the null hypothesis.

C. $t_{0.975}$ is the p value value for significance level $\alpha = 0.05$ and the 95% confidence interval is the rejection region for the null hypothesis.

D. None of the above

Answer: A.

## Question 21

Given the following data:

```r
x <- c(0.60, 0.52, 0.54, 0.09, 1.00, 0.05, 0.30, 0.26, 0.69, 0.29, 0.93)
y <- c(0.87, 0.93,  1.00,  0.07, -0.01,  0.20,  0.77,  0.57,  0.73,  0.74,  0.49)
```

We wish to apply simple linear regression to this data: $y = \beta_0 + \beta_1 x$. Then which of the following is correct?

A. The simple linear regression fit is quite good as the p value of $\beta_1$ is less than $0.05$.

B. The simple linear regression fit is quite bad, it can be improved by creating a model: $y = \beta_0 + \beta_1 x + \beta_2 z$, where $z = x^2$.

C. The simple linear regression fit is quite good as $R^2$ is close to 1.

D. None of the above

Answer: B.

```r
m <- lm(y ~ x)
summary(m)
```

```
## 
## Call:
## lm(formula = y ~ x)
## 
## Residuals:
##     Min      1Q  Median      3Q     Max 
## -0.6187 -0.2338  0.1394  0.2435  0.4183 
## 
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)  
## (Intercept)  0.55009    0.20952   2.625   0.0276 *
## x            0.05864    0.37063   0.158   0.8778  
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## 
## Residual standard error: 0.3689 on 9 degrees of freedom
## Multiple R-squared:  0.002774,   Adjusted R-squared:  -0.108 
## F-statistic: 0.02503 on 1 and 9 DF,  p-value: 0.8778
```

```r
z <- x**2
m <- lm(y ~ x + z)
summary(m)
```

```
## 
## Call:
## lm(formula = y ~ x + z)
## 
## Residuals:
##      Min       1Q   Median       3Q      Max 
## -0.13754 -0.08847  0.01024  0.06217  0.16332 
## 
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)    
## (Intercept) -0.11440    0.09486  -1.206    0.262    
## x            3.90959    0.42263   9.251 1.51e-05 ***
## z           -3.69388    0.39072  -9.454 1.29e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## 
## Residual standard error: 0.1121 on 8 degrees of freedom
## Multiple R-squared:  0.9181, Adjusted R-squared:  0.8976 
## F-statistic: 44.82 on 2 and 8 DF,  p-value: 4.505e-05
```

## Question 22

Consider two discrete random variables $X$ and $Y$ with the joint probability distribution $P(X = x, Y = y)$. The marginal probability $P(X = x)$ is defined as:

A. $P(X = x) = P(X = x, Y = y)$

B. $P(X = x) = \sum_x P(X = x|Y = y)$

C. $P(X = x) = \sum_x P(X = x, Y = y)$

D. $P(X = x) = \sum_y P(X = x, Y = y)$

Answer: D

## Question 23

You have a dataset called X_data in R containing a sample of numerical values. You want to test if this sample comes from a normal distribution using the Kolmogorov-Smirnov test. Which of the following R code snippets correctly performs this test?

A. ks.test(X_data, "pnorm")

B. ks.test(X_data, "dnorm")

C. t.test(X_data)

D. ks.test(X_data, "qnorm")

Answer: A

## Question 24

Consider the following R command:

```r
dbinom(2, 10, 0.4)
```

```
## [1] 0.1209324
```

Which of the following is correct?

A. It computes the probability density function's value at 2 of a Bernoulli distributed random variable.

B. It computes the cumulative distribution function's value at 2 of a Bernoulli distributed random variable.

C. It computes the probability density function's value at 2 of a Binomial distributed random variable.

D. It computes the cumulative distribution function's value at 2 of a Binomial distributed random variable.

Answer: C

## Question 25

Consider the following code to compute the test statistic in a one-sample two-sided Student's t test:

```r
x <- c(10, 10.5, 3.5, 9.7, 5.5)
n <- length(x)
mu0 <- 10

# Compute the test statistic
tobs <- (mean(x) - mu0) / (sd(x)/sqrt(n))
```

Which of the following is correct?

A. The null hypothesis (H0) is that the average value of the population is 0, tobs = -3.43.

B. The null hypothesis (H0) is that the average value of the population is 0, tobs = -1.53.

C. The null hypothesis (H0) is that the average value of the population is 10, tobs = -3.43.

D. The null hypothesis (H0) is that the average value of the population is 10, tobs = -1.53.

Answer: D

## Question 26

Experiment 1: Goal is to compare the taste and overall satisfaction of Brand A and Brand B coffee. Each participant will be given a cup of Brand A coffee and a cup of Brand B coffee. Participants will be asked to taste both coffees without knowing which brand they are drinking (blind taste test). After tasting, participants will fill out a survey rating each coffee on taste, aroma, and overall satisfaction.

Experiment 2: Goal is to compare two kinds of sleeping medicine A and B. 10 test persons try both kinds of medicine and the extra sleep hours obtained is measured.

Which of the following is correct?

A. 1 and 2 are both randomized experiments.

B. 1 is a randomized experiment and 2 is paired experiment.

C. 1 is a paired experiment and 2 is randomized experiment.

D. 1 and 2 are paired experiments.

Answer: D

## Question 27

Which of the following regarding confidence intervals for the mean is true?

*Consider the level of significance to be 0.05.*

A.The 99% confidence interval is wider than the 95% confidence interval.

B. 99% confidence interval uses 0.995 quantile while the 95% confidence interval uses the 0.975 quantile of the Student's t distribution with number of samples – 1 degrees of freedom.

C.The confidence intervals for the variance and standard deviation are computed using the $\chi^2$ (chi-squared) distribution.

D. All of the above

Answer: D

## Question 28

What is forecasting?

A. It is the process of determining the values of $\beta_0$ and $\beta_1$ for a linear model of the form $y = \beta_0 + \beta_1 x$, where $y$ is the output variable and $x$ is the input variable.

B. It is the process of determining the error of the linear model $y = \beta_0 + \beta_1 x$.

C. It is the process of determining the values of f(x_future) where x_future refers to future values (unseen values) of x, where f is a given model.

D. All of the above.

Answer: C

## Question 29

Which of the following represents a multiplicative model of a time series $z_t$?

*Here t is the time, $z_t$ is the observed time series, $m_t$ is the trend, $s_t$ is the seasonal effect, $\epsilon_t$ is the error term*

A. $z_t = m_t * s_t + \epsilon_t$

B. $z_t = m_t + s_t + \epsilon_t$

C. $z_t = m_t + \epsilon_t$

D. None of the above

Answer: A

## Question 30

Given P (Y = 0 | X = 5) = 1/3, P(Y = 0, X = 5) = 1/7, P(X = 5) = 3/7 and P(Y = 0) = 4/7, which of the following is true?

*Here P represents the probability, X and Y are two discrete random variables.*

A. X = 5 and Y = 0 are independent events because P (Y = 0, X = 5) = P(Y = 0) * P(X = 5)

B. X = 5 and Y = 0 are not independent events because P (Y = 0, X = 5) = P(Y = 0 | X = 5) * P(X = 5)

C. X = 5 and Y = 0 are not independent events because P (Y = 0 | X = 5) ≠ P(Y = 0)

D. B and C

Answer: D